Planning and Reinforcement Learning in AI Systems

(PARLAI)

Lesson 6b - Model-Free RL (Part 2)

Sarah Keren

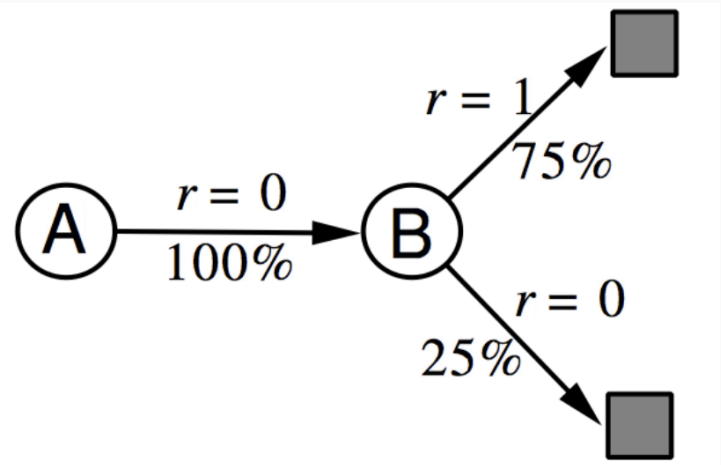
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Temporal-Difference Learning

AB Example

Two states A, B; no discounting; 8 episodes of experience

- A, 0, B, 0
- B, 1
- B, 1
- B, 1
- B, 1
- B, 1
- B, 1
- B, 0



What is $V(A)$, $V(B)$?

Temporal-Difference Learning
TD Evaluation
TD Control
MC vs. TD
Conclusion

From Monte Carlo to Temporal-Difference (TD) Learning

Temporal-difference (TD) learning has several advantages over Monte-Carlo (MC)

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Temporal-difference (TD) learning has several advantages over Monte-Carlo (MC)

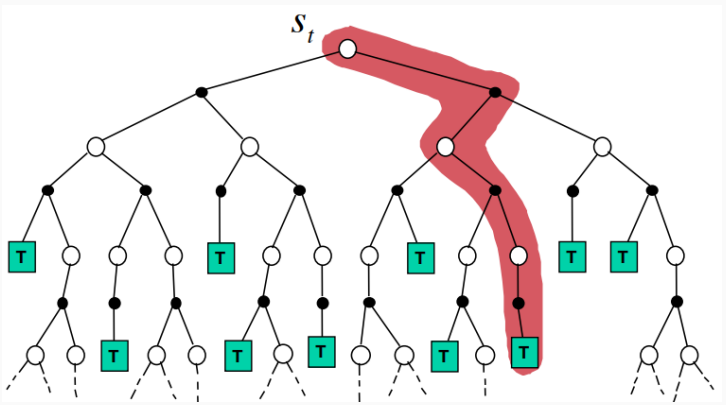
- Lower variance
- Online
- Works with incomplete sequences

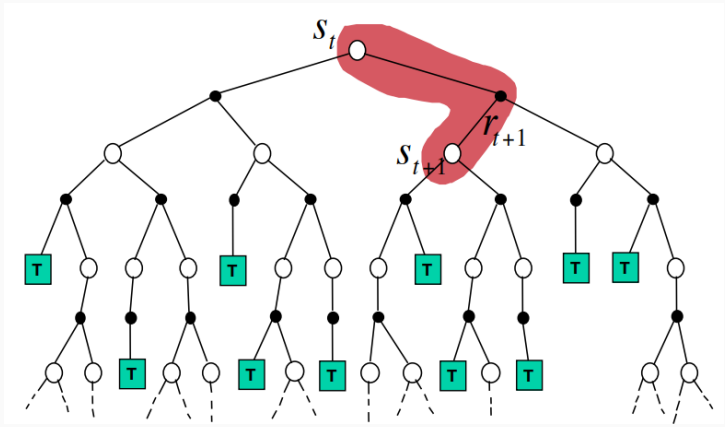
Natural idea: use TD instead of MC in our control loop

- Apply TD to $Q(s, a)$
- Use ϵ-greedy policy improvement
- Update every time-step.

From Monte Carlo to Temporal-Difference (TD) Learning

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Reminder: Bellman Formulas

Bellman equation (for a given deterministic policy):

Multiple variations, e.g.,

$$V_\pi(s) = R(s) + \gamma \sum_{s'} \mathcal{P}(s'|s, \pi(s)) \cdot V_\pi(s')$$

Bellman equation (for a given stochastic policy):

$$V_\pi(s) = \sum_a \pi(a \mid s) \left[R(s,a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s, a) V_\pi(s') \right]$$

Using Q

$$Q_\pi(s,a) = R(s,a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s, a) \sum_{a'} \pi(a' \mid s') Q_\pi(s', a')$$

Can we use these in RL?

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Reminder: Bellman Optimality Equations

$$V^*(s) = \max_a \left[R(s,a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s,a) V^*(s') \right]$$

$$Q^*(s,a) = R(s,a) + \gamma \sum_{s'} \mathcal{P}(s' \mid s,a) \max_{a'} Q^*(s',a')$$

$$V^*(s) = \max_a Q^*(s,a)$$

Can we use these in RL?

Reminder: during the learning process, we see (possibly partial) trajectories of the form:

$$\tau = s_0, a_1, r_1, s_1, \ldots, s_{T-1}, a_{T-1}, r_T, s_T$$

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TD Evaluation

Temporal-Difference (TD) Evaluation

- For each transition from s to s', we perform update

$$V_\pi(s) := V_\pi(s) + \alpha(R(s) + \gamma V_\pi(s') - V\pi(s))$$

with α as the learning rate

How does this move us closer to satisfying the Bellman constraint ?

$$V_\pi(s) = R(s) + \gamma \sum_{s'} \mathcal{P}(s'|s, \pi(s)) \cdot V_\pi(s')$$

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$$V_\pi(s) = R(s) + \gamma \sum_{s'} \mathcal{P}(s'|s, \pi(s)) \cdot V_\pi(s')$$

- $R(s) + \gamma V_\pi(s')$ is a (noisy) sample of utility based on the next state.
- The update maintains a "mean" of (noisy) utility samples
- How do we guarantee convergence to the true values ?

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Temporal-Difference (TD) Evaluation

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- $R(s) + \gamma V_\pi(s')$ is a (noisy) sample of utility based on the next state.
- The update maintains a "mean" of (noisy) utility samples
- How do we guarantee convergence to the true values ?
  - If the learning rate decreases appropriately with the number of samples (e.g. $\frac{1}{n}$), then the utility estimates will converge to true values (non-trivial).

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TD Control

Temporal-Difference (TD) Control

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- Simplest temporal-difference learning algorithm: $TD(0)$
  - Update value $V(s_t)$ toward estimated return $R_{t+1} + \gamma V(s_{t+1})$

$$V(s_t) \leftarrow V(s_t) + \alpha(R_{t+1} + \gamma V(s_{t+1}) - V(s_t))$$

- If we are using Q

$$Q(s_t, a_t) \leftarrow Q(s_t, a_t) + \alpha(R_{t+1} + \gamma Q(s_{t+1}, a_{t+1}) - Q(s_t, a_t))$$

Which policy do we use here?

Temporal-Difference (TD) Control

Do local updates of utility/value function on a per-action basis.

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Temporal-Difference (TD) Control

Do local updates of utility/value function on a per-action basis.

What does TD learn from?

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Temporal-Difference (TD) Control

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Do local updates of utility/value function on a per-action basis.

What does TD learn from? TD methods learn directly from episodes of experience

Model-free or model-based?

Temporal-Difference (TD) Control

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Do local updates of utility/value function on a per-action basis.

What does TD learn from? TD methods learn directly from episodes of experience

Model-free or model-based? TD is model-free: no learning of MDP transitions / rewards (doesn't try to estimate entire transition function).

Is there a need to wait for the episode to end?

Temporal-Difference (TD) Control

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Do local updates of utility/value function on a per-action basis.

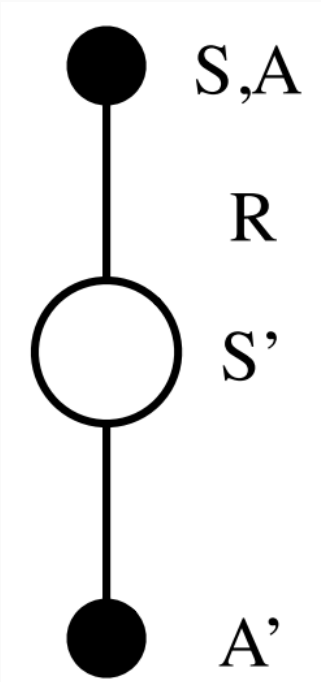
What does TD learn from? TD methods learn directly from episodes of experience

Model-free or model-based? TD is model-free: no learning of MDP transitions / rewards (doesn't try to estimate entire transition function).

Is there a need to wait for the episode to end? TD learns from incomplete episodes, by **bootstrapping** - updates a guess towards a guess.

On-Policy Control With SARSA

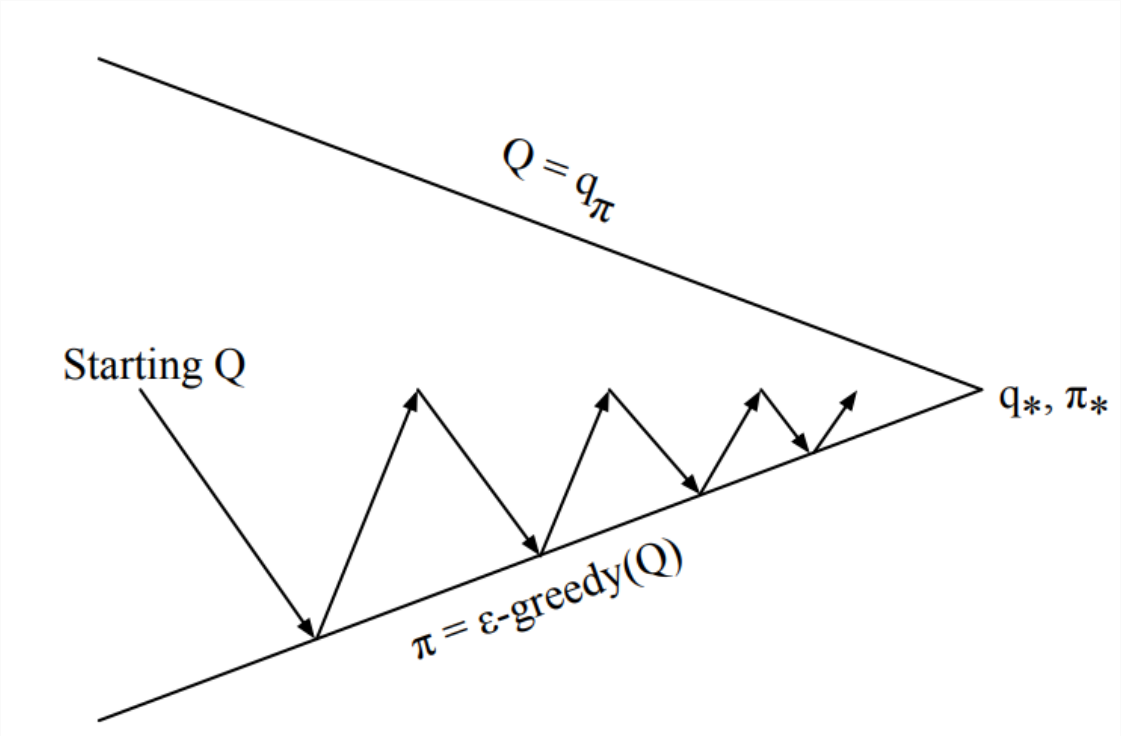
Updating action-value functions with



$$Q(S, A) \leftarrow Q(s, A) + \alpha(R + \gamma Q(S', A') - Q(S, A))$$

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On-Policy Control With SARSA



Every time-step:

- Policy evaluation: Sarsa $Q \approx Q_\pi$
- Policy improvement: ϵ-greedy improvement

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On-Policy Control With SARSA

SARSA (On-Policy TD Control)

Initialize:
$Q(s, a) \leftarrow$ arbitrary for all $s \in \mathcal{S}$, $a \in \mathcal{A}(s)$, $Q(\text{terminal}, \cdot) \leftarrow 0$
while true **do**
    Initialize start state S
    Choose action A using an ϵ-greedy policy derived from Q
    while S is not terminal **do**
        Execute action A, observe reward R and next state S'
        Choose action A' derived from Q (e.g., ϵ-greedy policy)

$$Q(S, A) \leftarrow Q(S, A) + \alpha \left(R + \gamma Q(S', A') - Q(S, A)\right)$$

        $S \leftarrow S'$
        $A \leftarrow A'$
    end while
end while

$$Q(S, A) \leftarrow Q(S, A) + \alpha(R + \gamma Q(S', A') - Q(S, A))$$

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Off-Policy Learning

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Evaluate **target policy** $\pi(a|s)$ to compute $V_\pi(s)$ or $Q_\pi(s,a)$ while following **behaviour policy** $\mu(a|s)$

$$\{s_1, a_1, R_2, \ldots, s_T\} \sim \mu$$

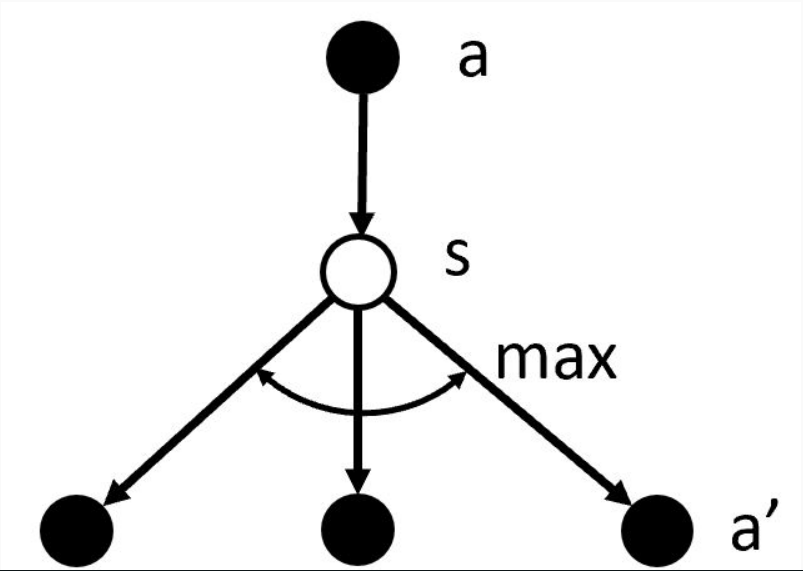
Why is this important?

- Learn from observing humans or other agents
- Re-use experience generated from old policies $\pi_1, \pi_2, ..., \pi_{t1}$
- Learn about optimal policy while following exploratory policy
- Learn about multiple policies while following one policy

Off-Policy Control with Q-Learning

- We now consider off-policy learning of action-values $Q(s, a)$
- Next action is chosen using the behaviour policy $A_{t+1} \sim \mu(|S_t)$ but we update the target policy by considering the possible actions that could be applied to the next state and choose one with maximal value.

$$Q(S, A) \leftarrow Q(S, A) + \alpha(R + \gamma \max_a Q(S', a) - Q(S, A))$$



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Off-Policy Control with Q-Learning

- In Q-learning we allow both behaviour and target policies to improve
- The target policy π is greedy w.r.t. $Q(s, a)$

$$\pi(s) = \arg\max_{a'} Q(s, a')$$

- The behaviour policy μ is (e.g.) ϵ-greedy w.r.t. $Q(s, a)$
- The Q-learning target then simplifies:

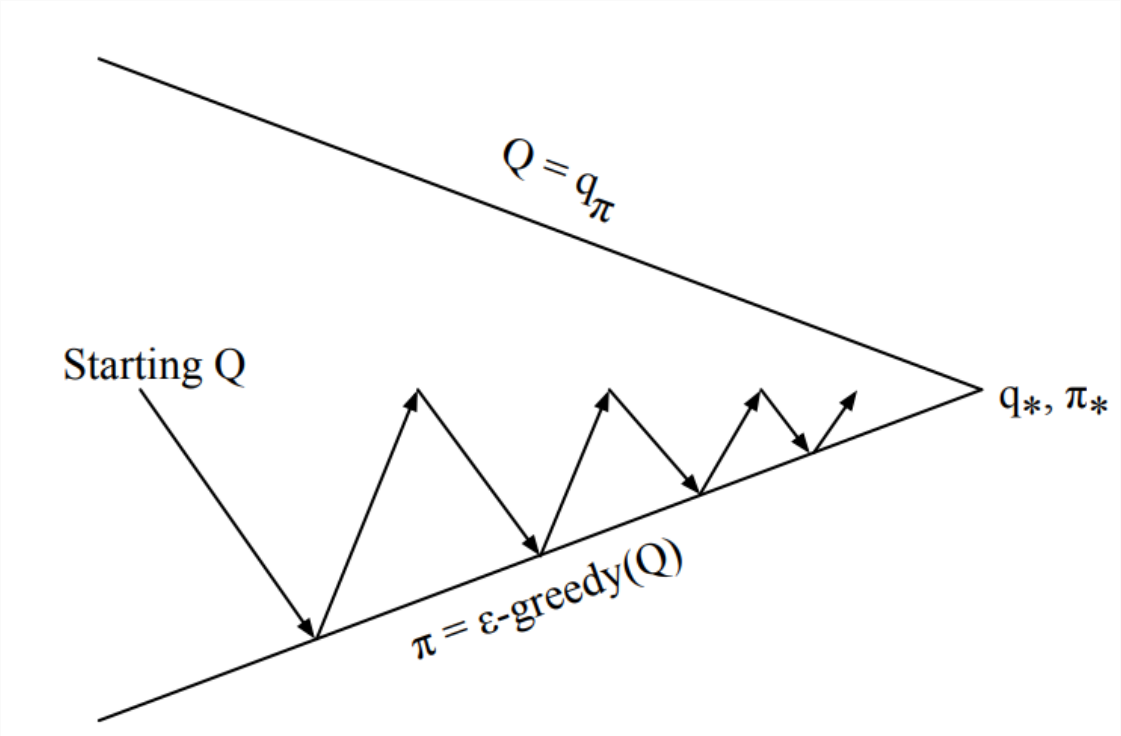
$$R + \gamma Q(s', a')$$

$$= R + \gamma Q(s', \arg\max_{a} Q(s', a))$$

$$= R + \max_{a} \gamma Q(s', a)$$

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Off-Policy Control With Q-Learning



Every time-step:

- Policy evaluation: Q-learning $Q \approx \epsilon$-greedy
- Policy improvement: ϵ-greedy improvement

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Q-Learning

Q-Learning (Off-Policy TD Control)

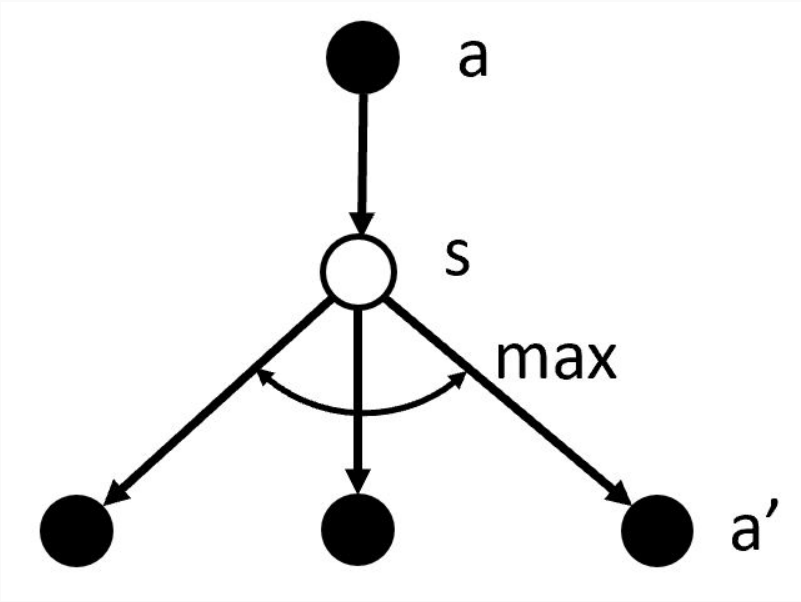
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while true **do**
    Initialize start state S
    while S is not terminal **do**
        Choose action A with ϵ-greedy policy derived from Q
        Execute action A
        Observe reward R and next state S'
        $Q(S, A) \leftarrow Q(S, A) + \alpha \left(R + \gamma \max_a Q(S', a) - Q(S, A)\right)$
        $S \leftarrow S'$
    end while
end while

$$Q(S, A) \leftarrow Q(S, A) + \alpha(R + \gamma \max_a Q(S', a) - Q(S, A))$$

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Off-Policy Control with Q-Learning

$$Q(S,A) \leftarrow Q(S,A) + \alpha(R + \gamma \max_a Q(S',a) - Q(S,A))$$



Theorem

Q-learning control converges to the optimal action-value function, $Q(s,a) \rightarrow Q^*(s,a)$

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SARSA vs. Q-Learning

SARSA:

$$Q(S_t, A_t) \leftarrow Q(S_t, A_t) + \alpha(R_{t+1} + \gamma Q(S_{t+1}, A') - Q(S_t, A_t))$$

Q-Learning:

$$Q(S, A) \leftarrow Q(S, A) + \alpha(R + \gamma \max_a Q(S', a) - Q(S, A))$$

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MC vs. TD

MC vs. TD

- Goal: learn and optimize value function online from experience
- Incremental every-visit Monte-Carlo
  - Update value $V(s_t)$ toward actual return G_t
    (typically, $G_t = R_{t+1} + \gamma R_{t+2} + ... + \gamma^{T-1} R_T$)

$$V(s_t) \leftarrow V(s_t) + \alpha(G_t - V(s_t))$$

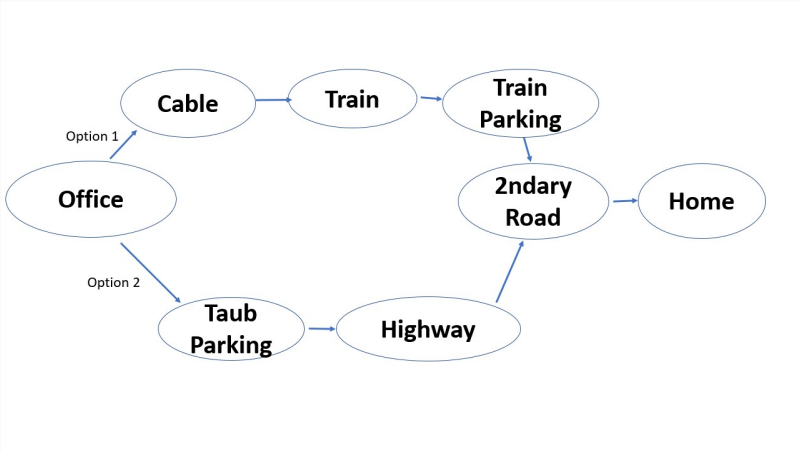
- Simplest temporal-difference learning algorithm: $TD(0)$
  - Update value $V(s_t)$ toward estimated return $R_{t+1} + \gamma V(s_{t+1})$

$$V(s_t) \leftarrow V(S_t) + \alpha(R_{t+1} + \gamma V(s_{t+1}) - V(s_t))$$

  - $R_{t+1} + \gamma V(s_{t+1})$ is called the **TD target**
  - $R_{t+1} + \gamma V(s_{t+1}) - V(s_t)$ is called the **TD error**

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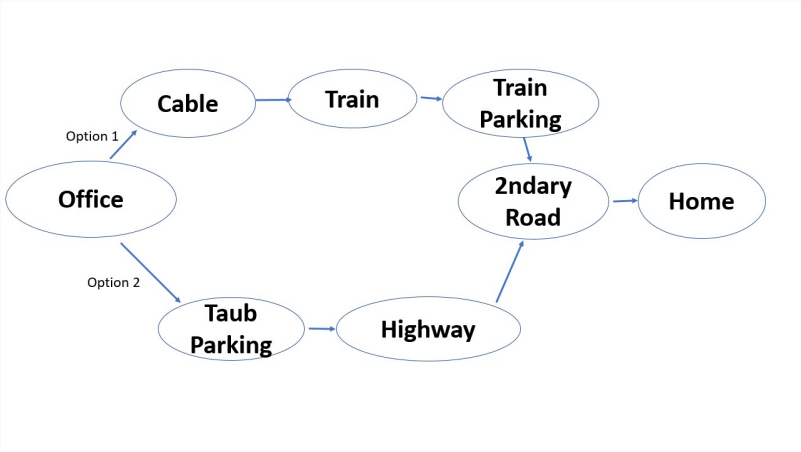
Going Home Example



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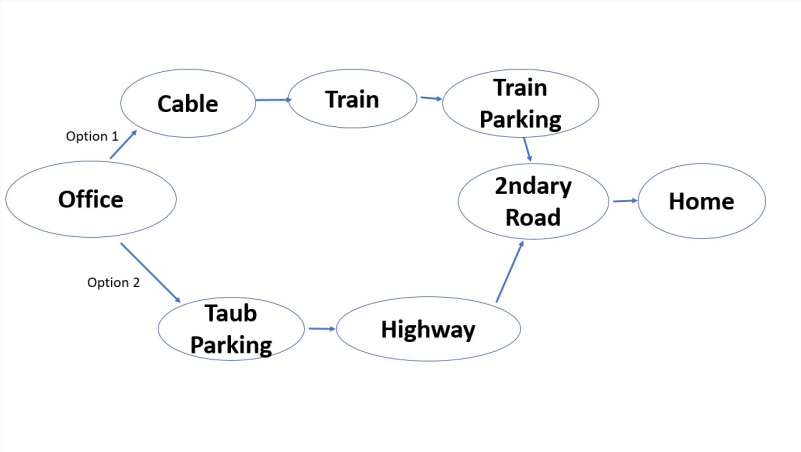


- τ_1 = office, 20, cable, 15, train, 30, train-park, 5, 2ndary, 15, home
- τ_2 = office, 20, cable, 15, train, 60, train-park, 5, 2ndary, 15, home
- τ_3 = office, 5, taub-park, 10, highway, 60, 2ndary, 15, home
- τ_4 = office, 5, taub-park, 10, highway, 60, 2ndary, 35, home
- τ_5 = office, 5, taub-park, 10, highway, 60

Going Home Example

$$V\pi(s_t) \leftarrow V\pi(s_t) + \alpha(G_t - V(s_t))$$
$$G_t = R_{t+1} + \gamma R_{t+2} + ... + \gamma^{T-1} R_T$$

$$V\pi(s_t) \leftarrow V\pi(s_t) + \alpha(R(s_t) + \gamma V\pi(s_{t+1}) - V\pi(s))$$
$$\gamma = 1 \; \alpha = 0.5$$



- τ_1 = office, 20, cable, 15, train, 30, train-park, 5, 2ndary, 15, home
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$$V_\pi(s_t) \leftarrow V_\pi(s_t) + \alpha(G_t - V(s_t))$$
$$G_t = R_{t+1} + \gamma R_{t+2} + ... + \gamma^{T-1} R_T$$
$$V_\pi(s_t) \leftarrow V_\pi(s_t) + \alpha(R(s_t) + \gamma V_\pi(s_{t+1}) - V_\pi(s))$$
$$\gamma = 1 \; \alpha = 0.5$$

		MC		TD	
State	0	1	2	1	2
Exit office	?	?	?	?	?
Enter Cable	?	?	?	?	?
Enter Taub Park	?	?	?	?	?
Enter Train	?	?	?	?	?
Enter Train Park	?	?	?	?	?
Enter Highway	?	?	?	?	?
Enter 2nd Road	?	?	?	?	?
Enter Home	?	?	?	?	?

- τ_1 = office, 20, cable, 15, train, 30, train-park, 5, 2ndary, 15, home
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MC vs. TD

- TD can learn before knowing the final outcome
  - TD can learn online after every step
  - MC must wait until end of episode before return is known
- TD can learn without the final outcome
  - TD can learn from incomplete sequences
  - MC can only learn from complete sequences
  - TD works in continuing (non-terminating) environments
  - MC only works for episodic (terminating) environments

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MC vs. TD

- Return $G_t = R_{t+1} + \gamma R_{t+2} + \cdots + \gamma^{T-1} R_T$ is an **unbiased** estimate of $V\pi(s_t)$
- True TD target $R_{t+1} + \gamma v_\pi(s_{t+1})$ is **unbiased estimate** of $V\pi(s_t)$.
- TD target $R_{t+1} + \gamma V(s_{t+1})$ is a **biased** estimate $V\pi(s_t)$. why?
- TD target has much lower variance than the return - why?

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- True TD target $R_{t+1} + \gamma v_\pi(s_{t+1})$ is **unbiased estimate** of $V\pi(s_t)$.
- TD target $R_{t+1} + \gamma V(s_{t+1})$ is a **biased** estimate $V\pi(s_t)$. why?
- TD target has much lower variance than the return - why?
    - Return depends on many random actions, transitions, rewards
    - TD target depends on one random action, transition, reward

MC vs. TD

- MC has high variance, zero bias
  - Good convergence properties
  - (even with function approximation)
  - Not very sensitive to initial value
  - Very simple to understand and use
- TD has low variance, some bias
  - Usually more efficient than MC
  - TD(0) converges to $V\pi(s)$
  - (but not always with function approximation)
  - More sensitive to initial value

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MC vs. TD

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- TD exploits Markov property
  - Usually more efficient in Markov environments
- MC does not exploit Markov property
  - Usually more effective in non-Markov environments

MC vs. TD

- MC converges to solution with minimum mean-squared error
  - Best fit to the observed returns

$$\sum_{k=1}^{K}\sum_{t=1}^{T_k}(G_t^k - V(s_t^k))^2$$

- $TD(0)$ converges to solution of max likelihood Markov model
  - Solution to the MDP $\langle \mathcal{S}, \mathcal{A}, \mathcal{P}, \mathcal{R}, \gamma \rangle$ that best fits the data

$$\mathcal{P}_{s,s'}^a = \frac{1}{N(s,a)}\sum_{k=1}^{K}\sum_{t=1}^{T_k} 1(s_t^k, a_t^k, s_{t+1}^k = s, a, s')$$

$$\mathcal{R}_s^a = \frac{1}{N(s,a)}\sum_{k=1}^{K}\sum_{t=1}^{T_k} 1(s_t^k, a_t^k = s, a) r_t^k$$

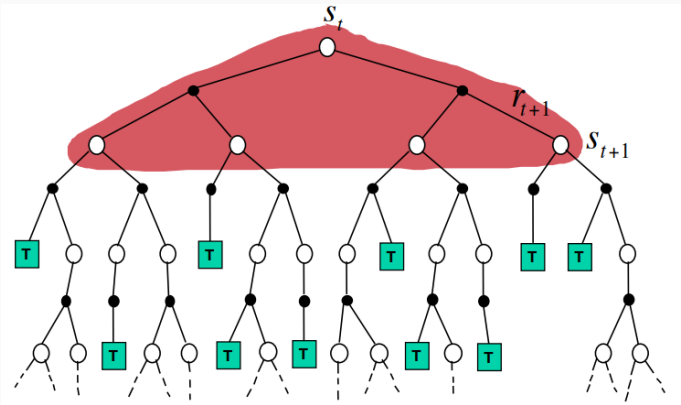
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Conclusion

Monte-Carlo vs. Temporal-Difference vs. Dynamic Programming Backup

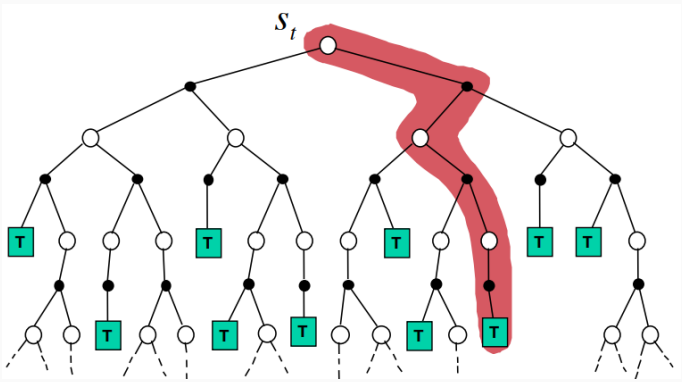
Which is which ?

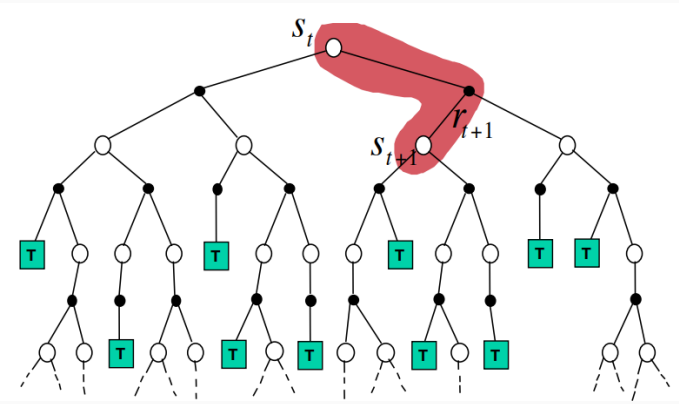
$$V(s_t) \leftarrow \mathbb{E}_\pi[R_{t+1} + \gamma V(s_{t+1})]$$



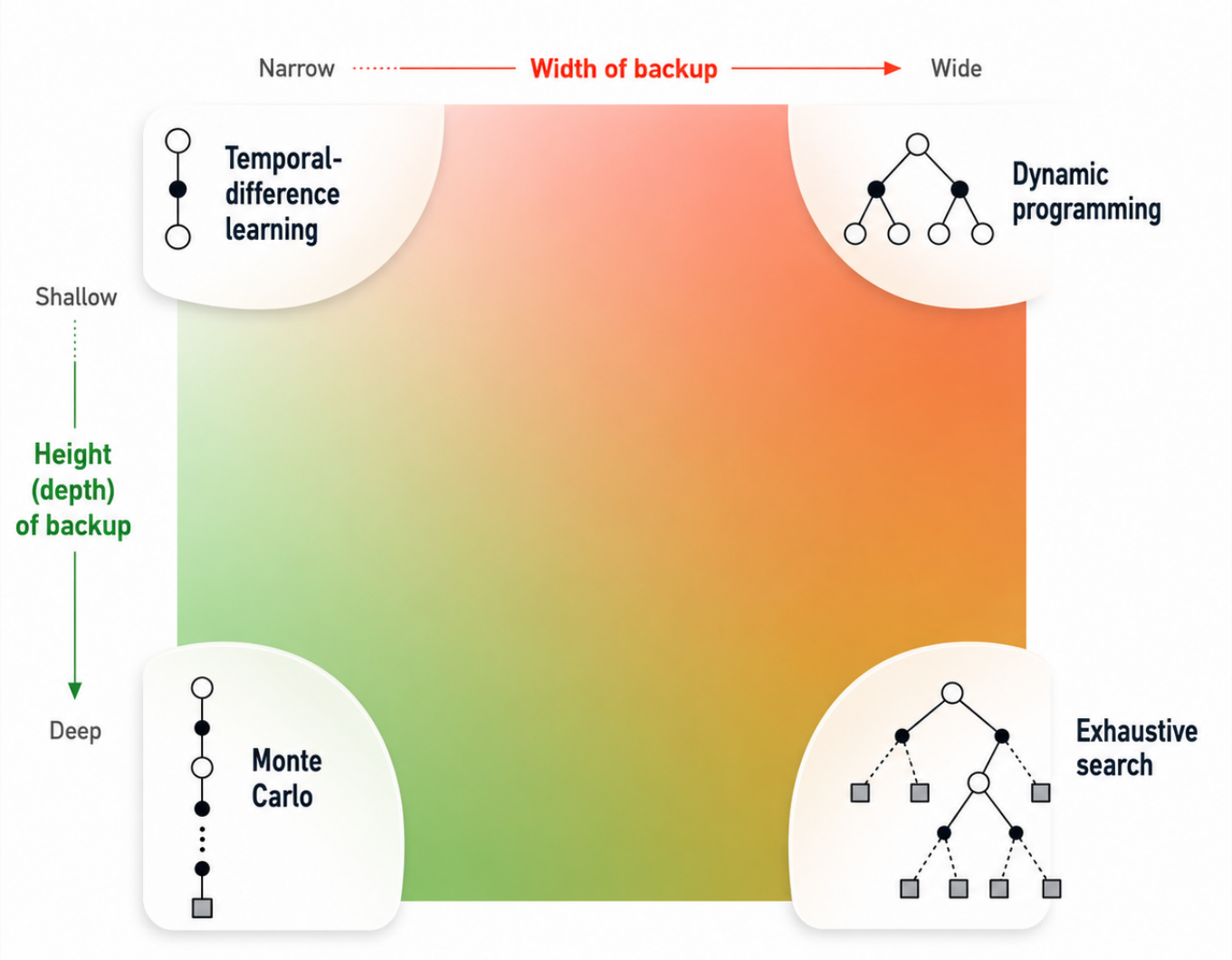
$$V(s_t) \leftarrow V(s_t) + \alpha(G_t V(s_t))$$

$$V(s_t) \leftarrow V(s_t) + \alpha(R_{t+1} + \gamma V(s_{t+1}) - V)]$$





Unified View of Reinforcement Learning



Temporal-Difference Learning
TD Evaluation
TD Control
MC vs. TD
Conclusion